1. Clinical Question & Core Hypothesis

Clinical Question

Tiered analysis highlights that organized local immunity (TLS/Immunoscore) should protect against NMIBC progression, yet some TLS-rich tumors still progress. Why does local immune organization fail in a subset of NMIBC? 2) Tier 1 paths used: I built primarily on CORE PATH 1 (SOX2 and MSH6) because it directly ties a stemness/transcriptional program (SOX2) and DNA-repair status (MSH6) to bladder prognosis; I also considered CORE PATH 2/3 (NDRG1, metabolic/host biomarkers) as contextual support but did not place them in the core claim to remain focused. 3) Tier 2 evidence that supports the mechanism: Immunoscore/immune-context paths justify using an imaging immune feature (TLS-like aggregates); NDRG1/DPYD and methylation markers (SDC2/TFPI2) suggest metabolic/epigenetic modifiers that could alter TLS functionality and are therefore listed as secondary modifiers. 4) Why PFI: For NMIBC the clinically critical outcome is progression to muscle-invasive disease (captured by progression-free interval); PFI is the most actionable and sensitive endpoint to test a mechanism where local immunity fails to prevent progression. 5) Deliberate exclusions: I excluded systemic metabolic biomarkers (HDL-C, ALP, TG, SHBG, IGF1) and cross-cancer markers (PSA, pZVR) from the core hypothesis because they broaden the scope beyond the central tumor–immune interaction; these remain in secondary_modifiers for exploratory analyses?

Working Hypothesis

In NMIBC, TLS-like lymphoid aggregates are present but do not guarantee improved progression-free interval (PFI) when tumors exhibit a SOX2-driven cancer stem cell program and high MSH6 expression (MMR-proficient). The proposed mechanism is twofold: (1) SOX2-induced stemness alters the tumor secretome and stromal remodeling in a way that impairs TLS maturation and function, reducing T cell activation within TLS niches (e.g., decreased CXCL13/T follicular helper signaling, increased immunosuppressive cytokines such as TGF-β, and dysfunctional dendritic cell education), thereby transforming TLS from an active anti-tumor microenvironment into a dysfunctional or exhausted structure. (2) High MSH6 expression results in proficient DNA repair and a relatively lower tumor mutational burden/neoantigen quality, diminishing TLS-driven neoantigen presentation and cytotoxic T cell priming within TLS. The combination—SOX2-high and MSH6-high in TLS-rich tumors—creates a high-risk subset in which TLS presence is insufficient to prevent progression, yielding shorter PFI despite TLS localization. This model accommodates the observed clinical heterogeneity and provides testable predictions about TLS maturation state and immune activity beyond mere TLS presence. Secondary modifiers (NDRG1, DPYD, PDIA2, SDC2, TFPI2) are anticipated to modulate the degree of TLS maturation and immune exclusion, thereby refining risk stratification within TLS-rich tumors. Testable predictions include: (a) among TLS-positive NMIBC, SOX2-high/MSH6-high tumors show shorter PFI than TLS-positive SOX2-low or MSH6-low tumors; (b) TLS maturation markers (e.g., CXCL13, CD21, BTNL2) and functional T cell signatures are reduced in SOX2-high/MSH6-high tumors; (c) higher expression of secondary modifiers correlates with impaired TLS maturation and worse PFI within TLS-positive, SOX2-high/MSH6-high cohorts; (d) neoantigen burden and TCR clonality patterns align with MSH6 status and TLS functionality. Feasible testing approaches are outlined below.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	SOX2, MSH6
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

SOX2: SOX2 and MSH6) because it directly ties a stemness/transcriptional program (SOX2) and DNA-repair status (MSH6) to bladder prognosis; I also considered CORE PATH 2/3 (NDRG1, metabolic/host biomarkers) as contextual support but did not place them in the core claim to remain focused.

MSH6: MSH6) because it directly ties a stemness/transcriptional program (SOX2) and DNA-repair status (MSH6) to bladder prognosis; I also considered CORE PATH 2/3 (NDRG1, metabolic/host biomarkers) as contextual support but did not place them in the core claim to remain

focused.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	NDRG1	Supporting modifier from knowledge graph
Secondary	DPYD	Supporting modifier from knowledge graph
Secondary	PDIA2	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: TLS presence is commonly associated with anti-tumor immunity, but TLS maturation and function—not just presence—predict clinical benefit;

Prediction 2: cancer stem cell programs (SOX2) rewire tumor-immune interfaces and secretome, often promoting stromal remodeling and immunosuppression;

Prediction 3: MMR proficiency (MSH6-high) tends to yield lower mutation/neoantigen burden, potentially dampening TLS-driven antigen-specific responses. In NMIBC, reports linking high MSH6 to worse outcomes support a paradox where proficient repair and lower neoantigen generation, coupled with stemness-driven immunomodulation, yields TLS-rich yet ineffective anti-tumor immunity.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- NDRG1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- DPYD: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.